Cat Coat Phenotypes and Survey Concordance

Colin Klick



Data used for this presentation

- Phenotypic survey data and genetic data from ~1500 cats through Darwin’s Ark

Darwin’s Ark - Elinor’s Idea

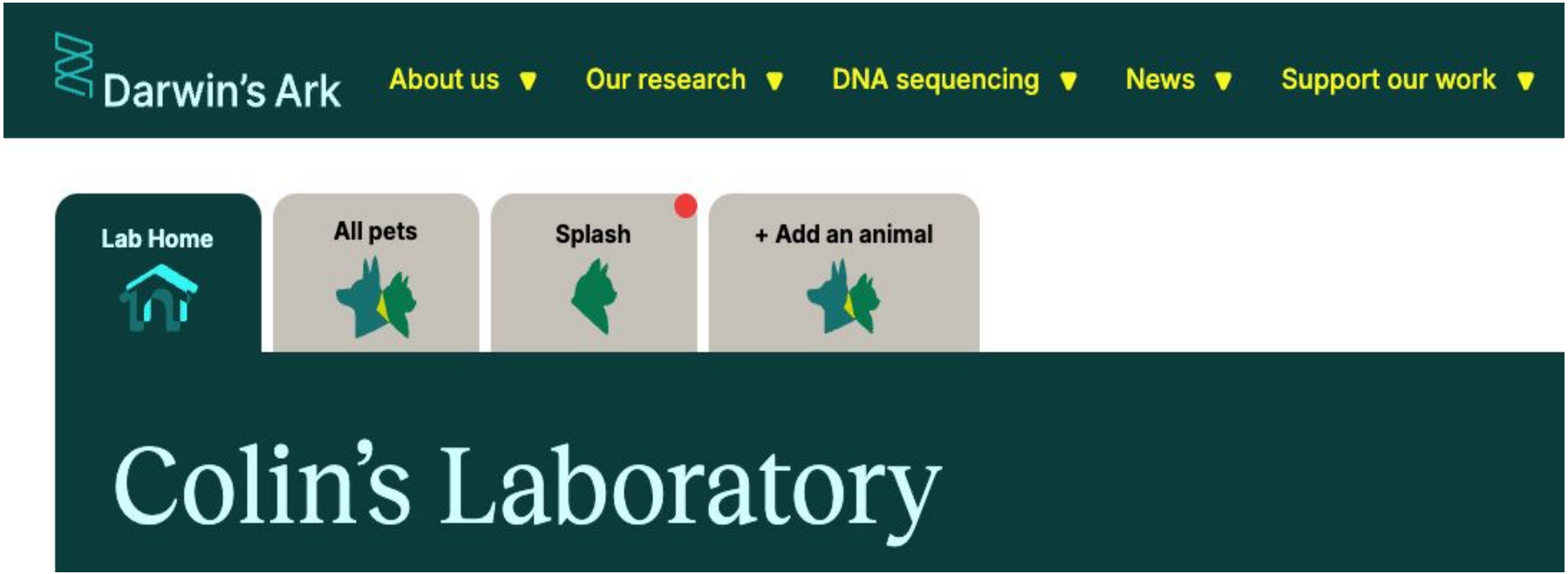
- Everyone wanted to tell Elinor about their dog
- The idea was to capture all this information to a format usable for science

A powerful model. For instance . . .

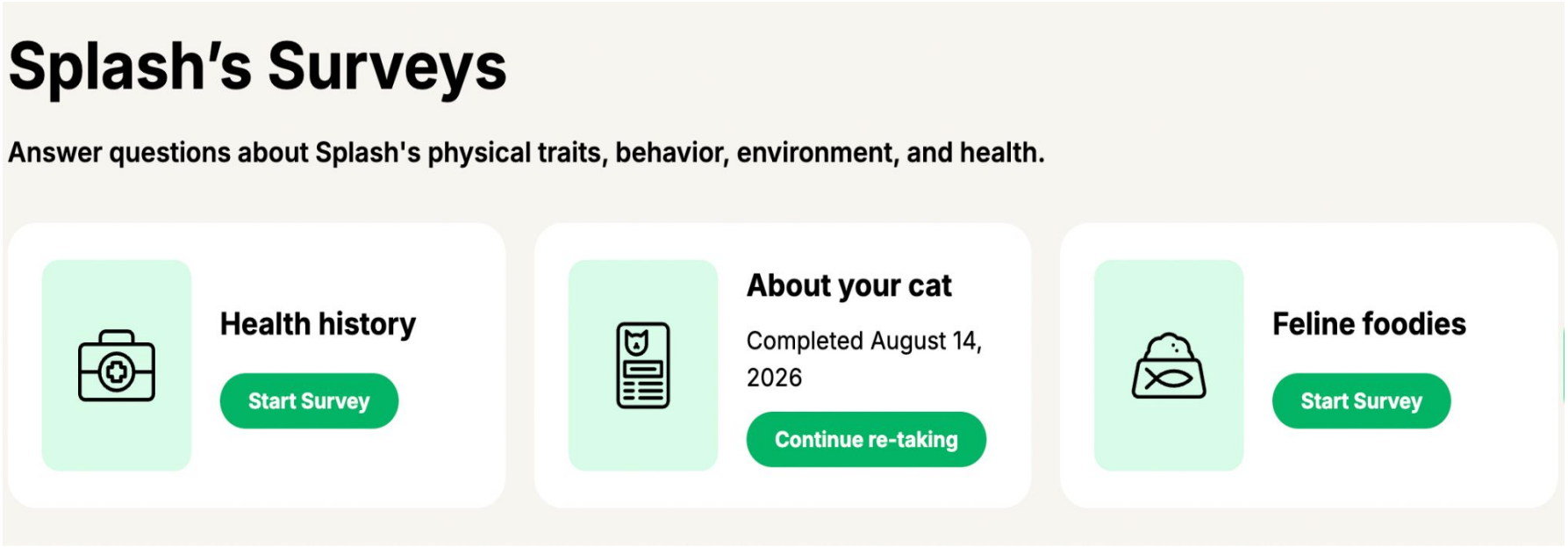
- Compulsive behaviors

Darwin’s Ark, a community science nonprofit

“Together, we can unlock the mysteries of pet behavior and genetics to help cats, dogs, and humans live healthier, happier lives.”



~260 Questions across 21 surveys



We are going to examine the coat and fur survey

Does Splash show a tabby pattern on their fur?

Tabbies have an M-shaped marking on their forehead, stripes by their eyes and back, legs, and tail, and some have patterns on the body. When answering, please just consider the non-white portions of Splash's coat.

- ○  **Tabby:** Distinctive M-shaped marking on forehead
- ○  **Not tabby**
- ○ ? **Not sure**

Why the coat and fur survey?

(1) Start by examining simpler traits caused by one or a few variants
(2) It’s the toughest survey in terms of owner accuracy, so it’s perfect for concordance/discordance analysis

Let’s dive in to a trait

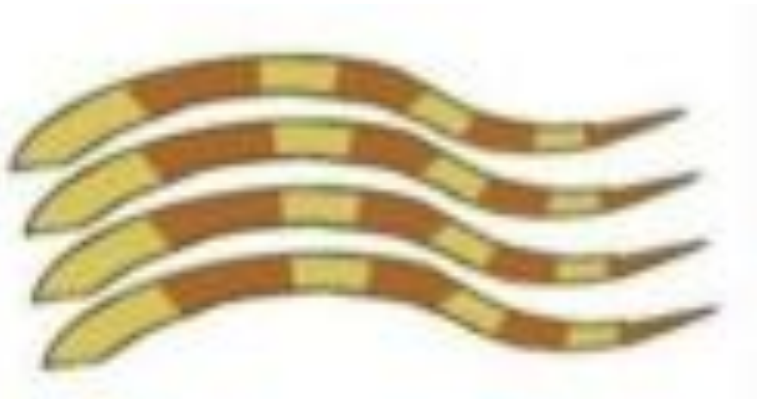




Normal agouti hairs

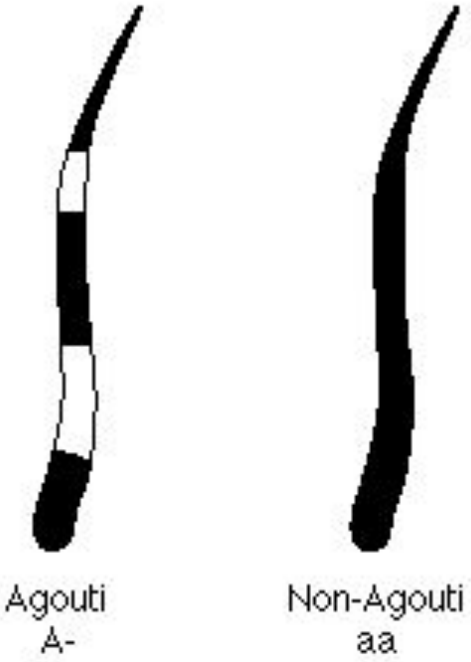


Eumelanin bands



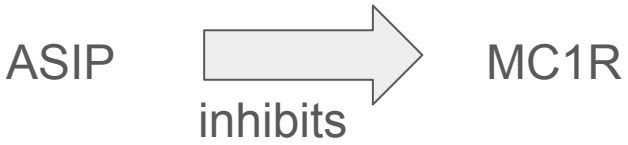
Phaeomelanin bands

Non-agouti hair (right)





ASIP turns the MC1R protein on and off



Eizirik et al. 2003 - Finding ASIP

- ASIP was a candidate gene from the literature (e.g., from mice)
- Used markers next to ASIP in an 89-cat pedigree
- Sequenced ASIP and found a 2 bp deletion, frameshifting the protein and truncating its critical C-terminal domain
- Confirmed with association with ~140 cats (small by today's standards)

Other variants affecting cat coat phenotypes

- Similarly to the non agouti variant, cat coat phenotypes are caused by one or a small number of variants

Previous studies used a combination of:

- Pedigree analysis
- Small-scale association analysis
- Development biology

Darwin’s Ark takes a different approach

- Population-scale, not pedigree-based
- Unbiased genome-wide scan, not candidate genes
- Owner-reported phenotypes, not clinical/expert scoring

A look into other traits and their variants

KIT - White spotting

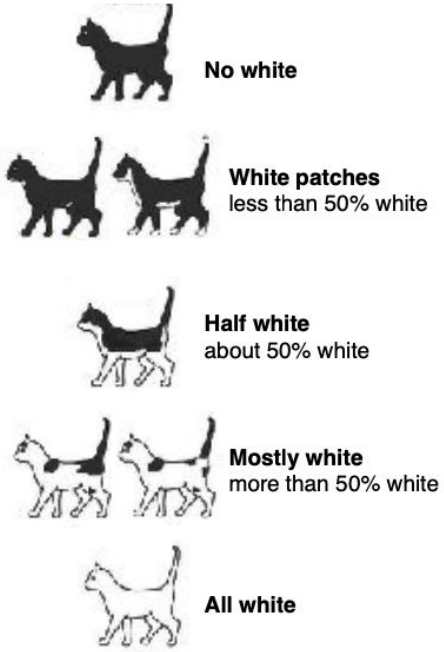
- Epistatic to other coat phenotypes

White

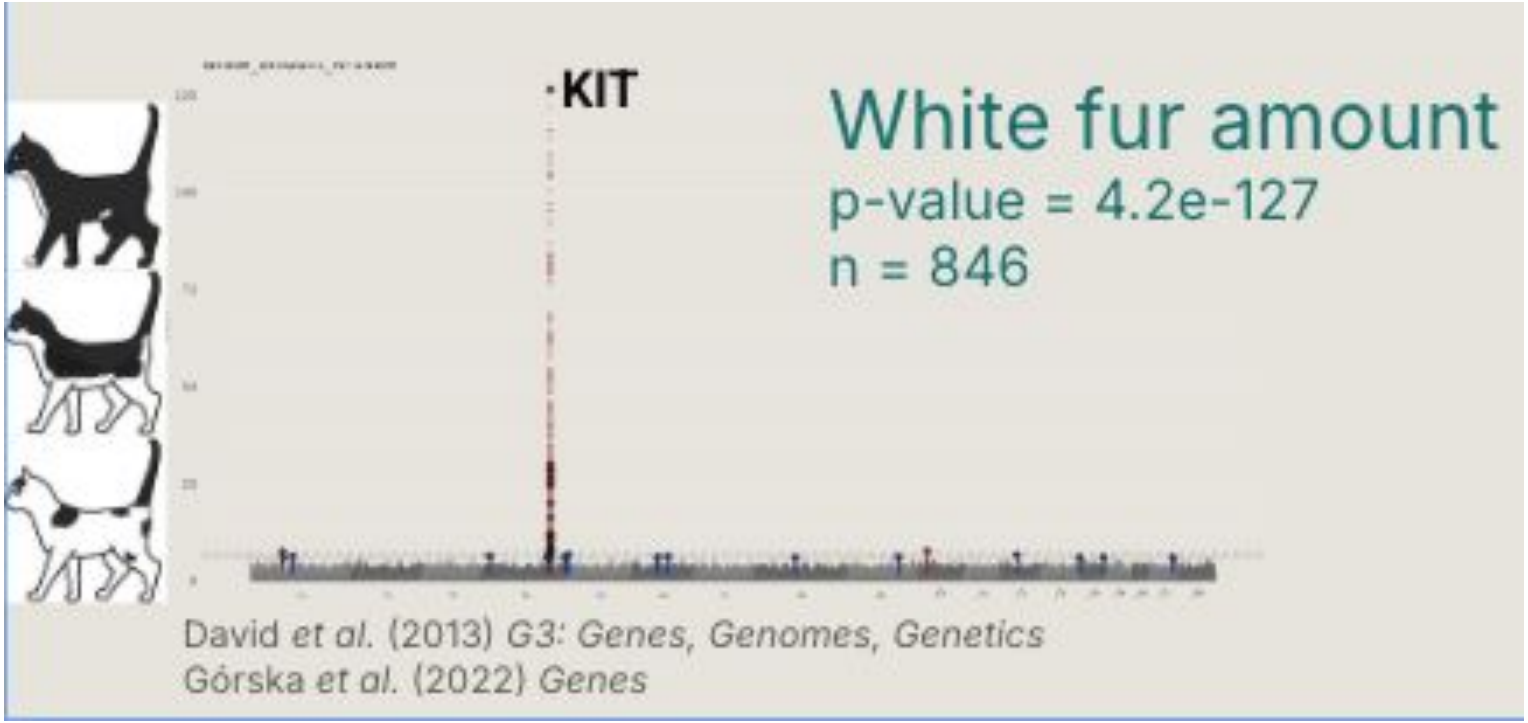
Does CAT have any **white fur**? Which image best captures the amount of white on CAT? (select one)

Please ignore other colors, such as cream or ivory, and any other patterns on CAT when answering this question.

- **No white**
- **White patches**
- **Half white**
- **Mostly white**
- **All white**
- **None of these**



Darwin's Ark GWAS replicates the association



Orange Locus

- For areas on coat that are not white, orange is epistatic to black

IF NOT white & IF NOT tortie & NOT tabby

CAT's coat color is most similar to which of the following: (select one)

Please ignore any white areas, and just consider CAT's non-white coat color when answering this question.

- **Black**
- **Chocolate**
- **Cinnamon**
- **Red**
- **Apricot**
- **Gray, Beige, or Cream**



Black: Jet-black or very dark brown, also called ebony



Chocolate: Medium-dark brown



Cinnamon: Milk-chocolate (reddish) colour, or a light brown



Red: Rich ginger red

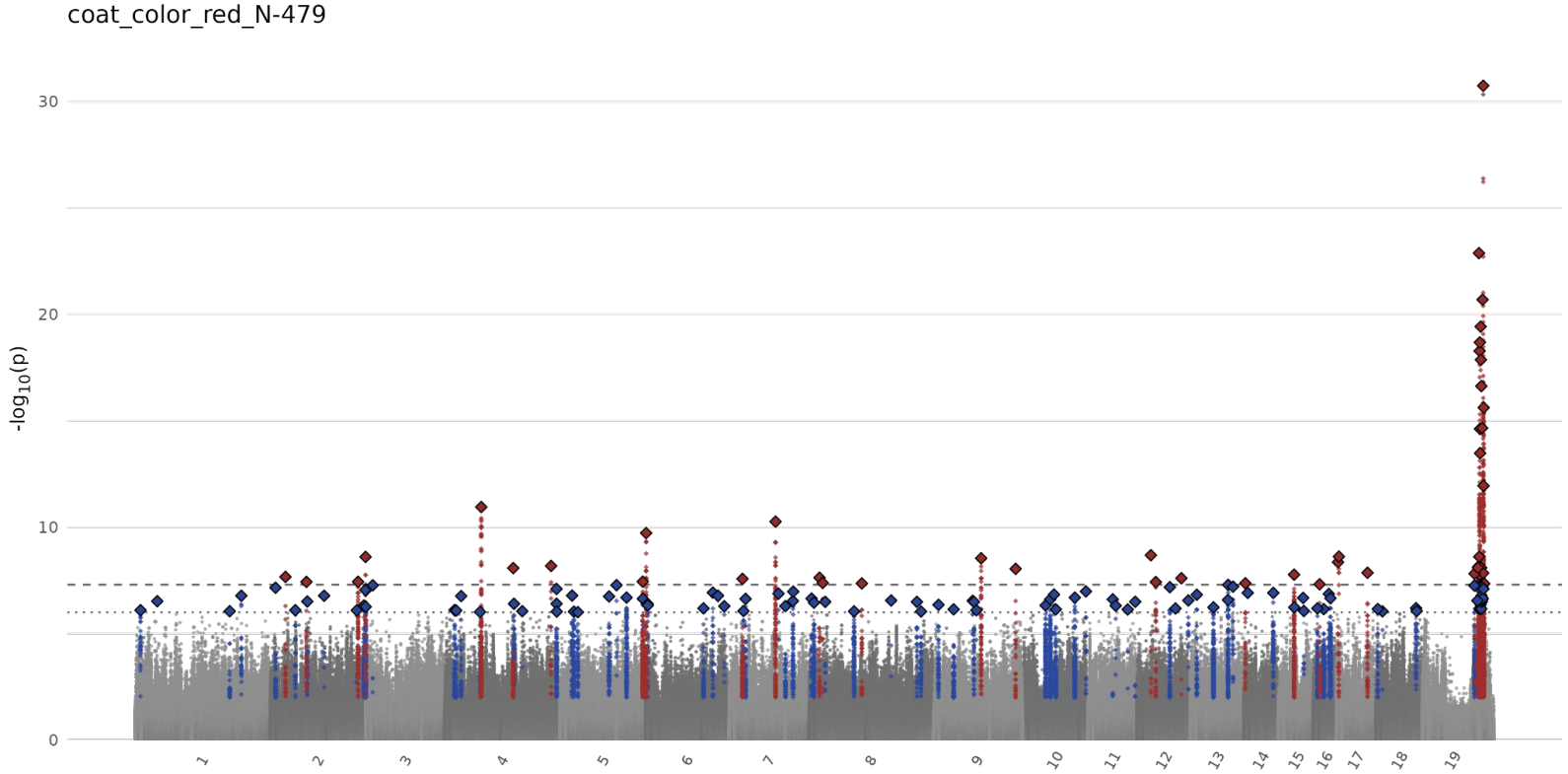


Apricot: Pink-brown or hot cream, with possibly a metallic sheen



Gray, Beige, or Cream: Any shade of gray, light gray, cream or caramel

Darwin’s Ark GWAS replicates the orange locus



Colorpoint

- Colorpoint variant acts independently of other traits

IF NOT white: Colorpoint

Does CAT show a **colorpoint** pattern?

Colorpoint cats have dark colors on the face, ears, legs, and tail, with a lighter version on the rest of the body.

- **Yes**
- **No**
- **Not sure**



Colorpoint cats have darker colors on the face, ears, legs, and tail

Dilute

- Dilute variant acts independently of other traits

IF gray or cream

CAT's coat color is most similar to which of the following: (select one)

Please ignore any white areas, and just consider CAT's non-white coat color when answering this question.

- **Blue**
- **Lilac**
- **Fawn**
- **Caramel**
- **Cream**



Blue: Any shade of cold-toned gray.



Lilac: A pale, warm-toned gray, like a warm blue-brown or dove gray.



Fawn: Hot cream colour, a lighter version of lilac.



Caramel: Cafe-au-lait colour or biscuit colour.



Cream: Buff colour.

Agouti/nonagouti (Tabby status)

- Describes the individual hairs on a cat

IF NOT white: Tabby

Does CAT show a **tabby** pattern on their fur?

Tabbies have M-shaped marking on their forehead, stripes by their eyes and along their back, legs, and tail, and some have patterns on the body. Please just consider the non-white portions of CAT's coat.

- **Yes**
- **No**
- **Not sure**



Distinctive M-shaped marking on forehead

If tabby, what is tabby type

IF Tabby

Which **tabby pattern** does CAT's non-white fur most closely resemble? (select one)

- **Classic tabby**
- **Mackerel tabby**
- **Spotted tabby**
- **Ticked tabby**
- **Rosetted tabby**
- **None of these**



Classic Tabby: a common "blotched" tabby pattern with dark stripes down the length of the back and dark swirls (bullseye) on sides of the body.



Mackerel Tabby : stripes of vertical unbroken thin lines instead of swirls.



Spotted Tabby : Vertical bars of colour are broken up into spots on the body. Stripes on leg, tail and face.



Ticked Tabby: No tell-tale stripes, swirls, or spots, but each individual hair is "ticked", or has the agouti pattern of bands of color along the hair shaft. May have stripes on face, legs and tail.



Rosetted Tabby: Clusters of spots; the centre of each cluster should be deeper version of background colour, found commonly in hybrid cats.

Describing a coat phenotype

- Colorpoint / Not colorpoint (TYR)
- Dilute / Not dilute (MLPH)
- Agouti / Not agouti
  - If Agouti: Mackerel / Blotched / Ticked / Other

Pictures - Rule-based organizing into categories



Not dilute, not colorpoint, non agouti (536)





Not dilute, not colorpoint, mackerel (322)







Variants from the literature (OMIA)

```
site_names = {
    ('chrD1', 44013031): 'siamese',
    ('chrD1', 44022819): 'burmese',
    ('chrA3', 24831082): 'agouti',
    ('chrA1', 94477275): 's59x',
    ('chrA1', 94536796): 'w841x',
    ('chrC1', 218197447): 'dilute',
    ('chrB1', 40428846): 'a18v',
    ('chrB1', 40429492): 'c63y',
    ('chrD4', 38142265): 'chocolate',
    ('chrD4', 38130163): 'cinnamon'
}
```

Building phenotype table from genotype

- Bcftools used to return genotype per site per cat
- Map genotype to allele count
- Pivot so each cat has its own row with allele counts for all sites

```
geno_wide.head()
```

site	sample	a18v	agouti	burmese	c63y	chocolate	cinnamon	dilute	s59x	siamese	w841x
0	BA00101	0.0	1.0	0.0	0.0	2.0	0.0	2.0	0.0	0.0	0.0
1	BA00102	0.0	1.0	0.0	0.0	2.0	0.0	2.0	0.0	0.0	0.0
2	BA00103	0.0	0.0	0.0	0.0	2.0	0.0	0.0	0.0	1.0	2.0
3	BA00104	0.0	1.0	0.0	0.0	2.0	0.0	0.0	0.0	1.0	1.0
4	BA00105	0.0	2.0	0.0	0.0	2.0	0.0	0.0	0.0	1.0	0.0

Using rules from the literature, assign phenotype

- Example: To classify the tabby pattern
  - If homozygous at one variant or heterozygous at two variants, then classify as ‘Blotched’
  - This designation can be overridden by ‘Ticked’ alleles

Classify cats according to their genotype

colorpoint_class	agouti_class	tabby_type	dilute_class	ticked_class	tyrp1_class	tabby_status
not_colorpoint	agouti	mackerel	dilute	not_ticked	black_at_tyrp1	tabby
not_colorpoint	agouti	mackerel	dilute	not_ticked	black_at_tyrp1	tabby
not_colorpoint	agouti	blotched	not_dilute	not_ticked	black_at_tyrp1	tabby
not_colorpoint	agouti	mackerel	not_dilute	not_ticked	black_at_tyrp1	tabby
not_colorpoint	non_agouti	NaN	not_dilute	not_ticked	black_at_tyrp1	not_tabby

Classify cats according to their owner's survey

barcode	survey_colorpoint	survey_agouti	survey_dilute	survey_tabby_status	survey_tabby_type
BA00101	not_colorpoint	agouti	NaN	tabby	ticked
BA00102	not_colorpoint	agouti	dilute	tabby	mackerel
BA00104	not_colorpoint	agouti	not_dilute	tabby	spotted
BA00105	NaN	NaN	NaN	NaN	NaN
BA00106	colorpoint	agouti	not_dilute	tabby	mackerel

Colorpoint

- Colorpoint variant acts independently of other traits

IF NOT white: Colorpoint

Does CAT show a **colorpoint** pattern?

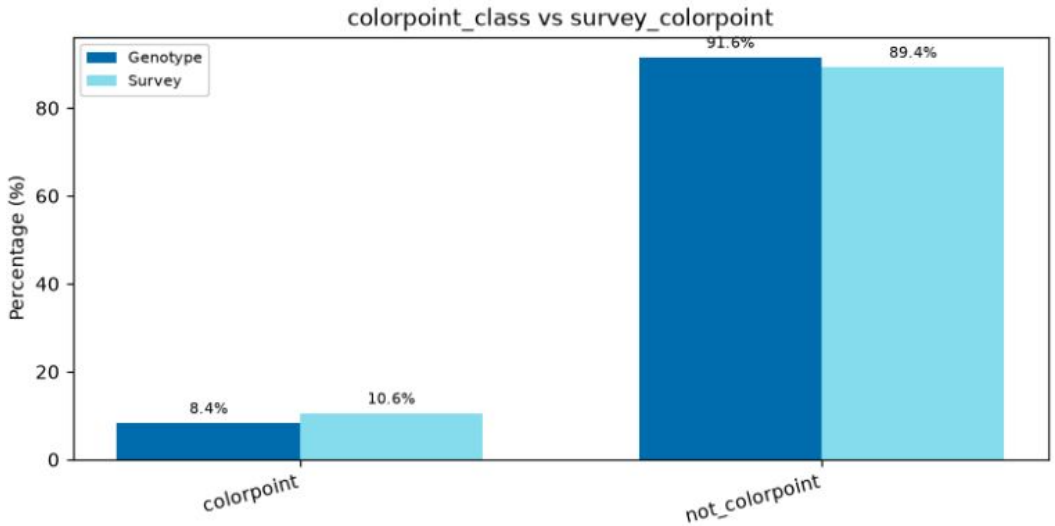
Colorpoint cats have dark colors on the face, ears, legs, and tail, with a lighter version on the rest of the body.

- **Yes**
- **No**
- **Not sure**



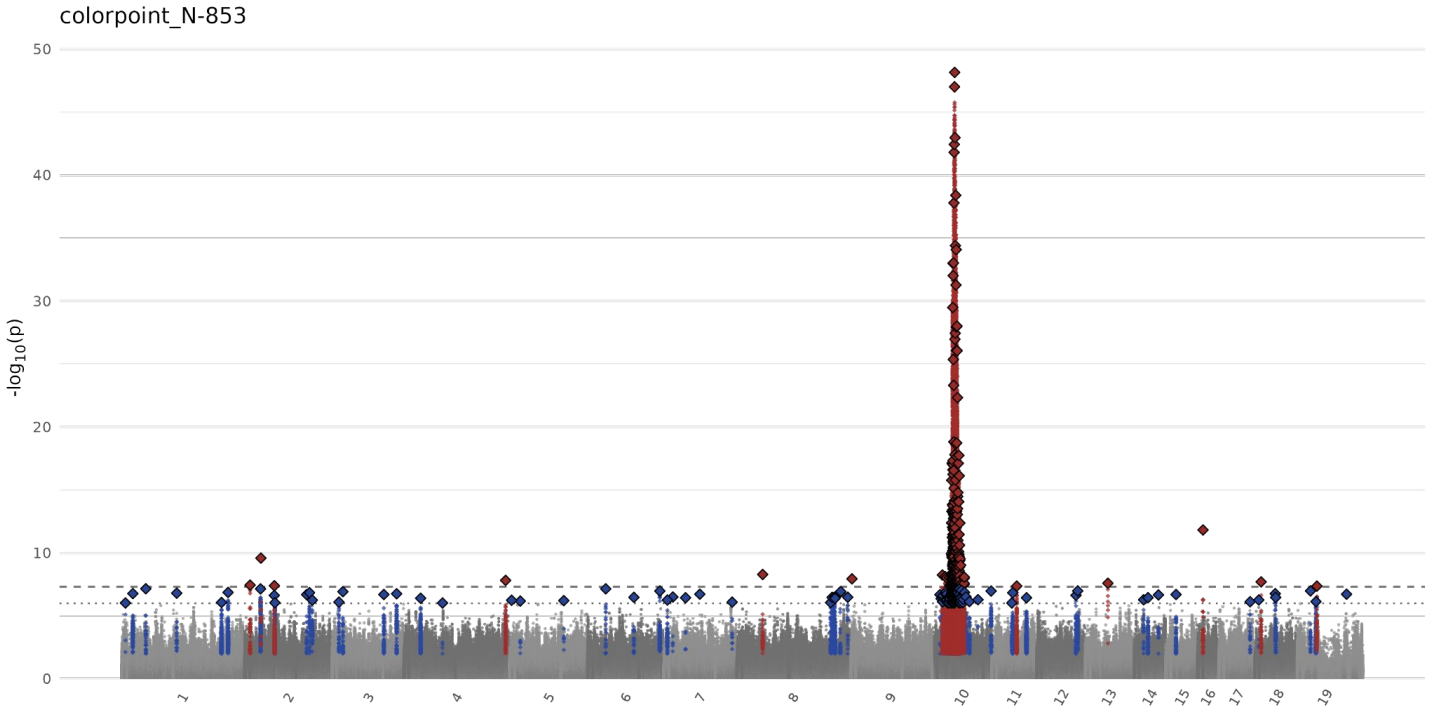
Colorpoint cats have darker colors on the face, ears, legs, and tail

Colorpoint



Overall discordance: 3.8%

Darwin’s Ark colorpoint GWAS



Dilute

- Dilute variant acts independently of other traits

IF gray or cream

CAT's coat color is most similar to which of the following: (select one)

Please ignore any white areas, and just consider CAT's non-white coat color when answering this question.

- **Blue**
- **Lilac**
- **Fawn**
- **Caramel**
- **Cream**



Blue: Any shade of cold-toned gray.



Lilac: A pale, warm-toned gray, like a warm blue-brown or dove gray.



Fawn: Hot cream colour, a lighter version of lilac.

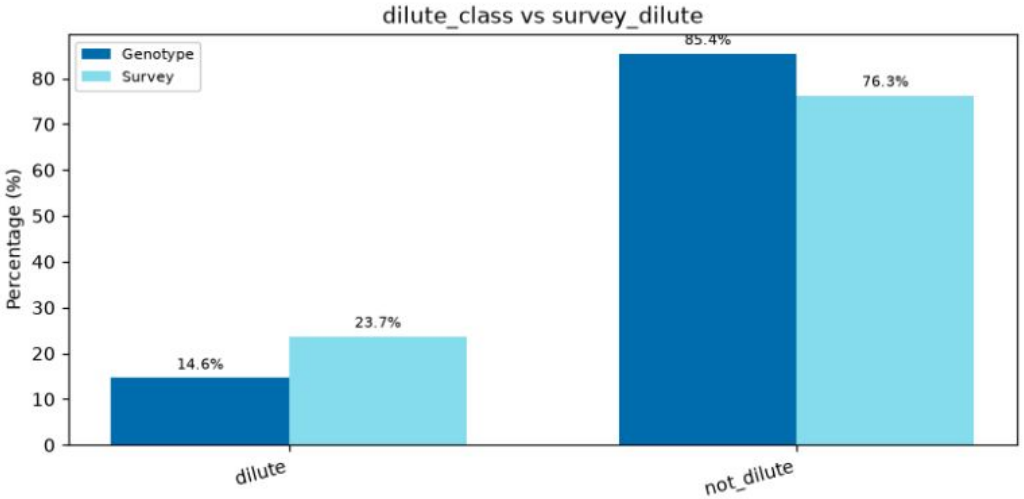


Caramel: Cafe-au-lait colour or biscuit colour.



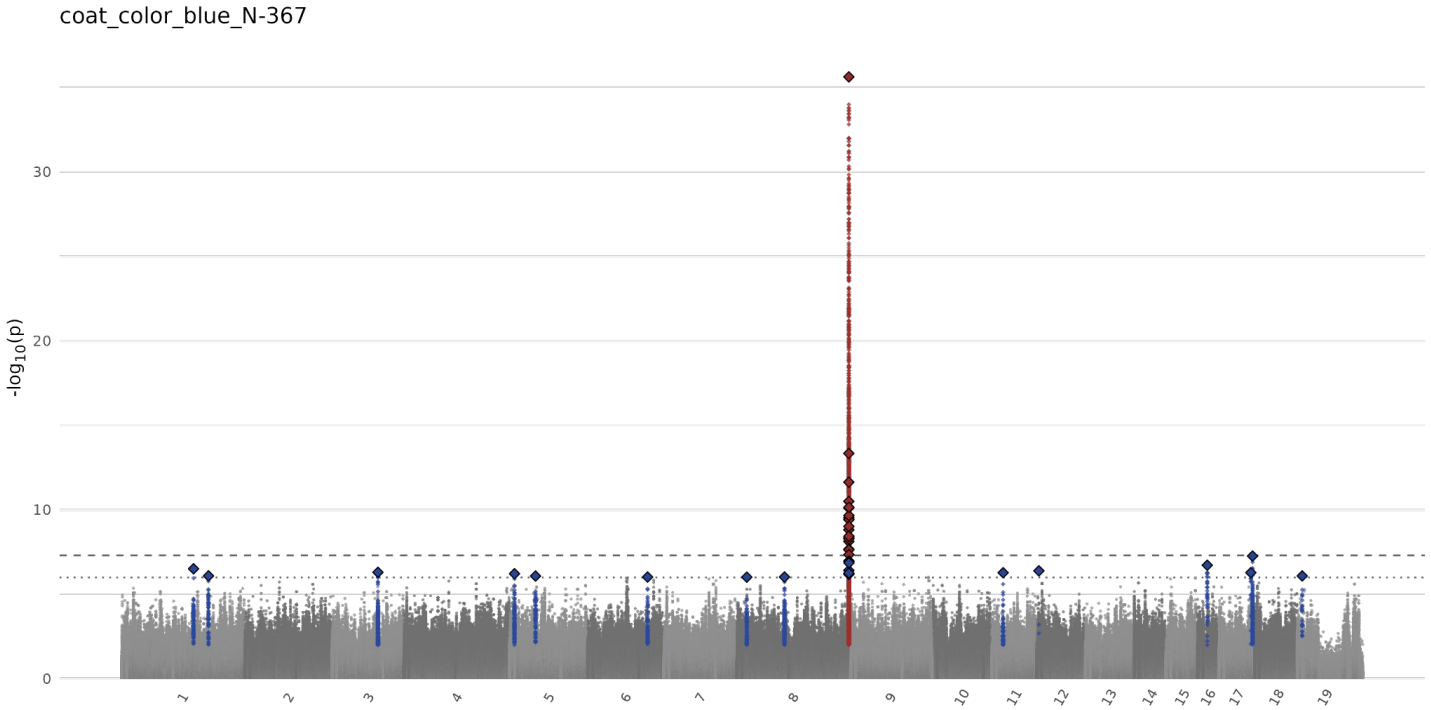
Cream: Buff colour.

Dilute Concordance



Overall discordance: 14.0%

Darwin’s Ark dilute GWAS (blue)



Agouti/nonagouti (Tabby status)

- Describes the individual hairs on a cat

IF NOT white: Tabby

Does CAT show a **tabby** pattern on their fur?

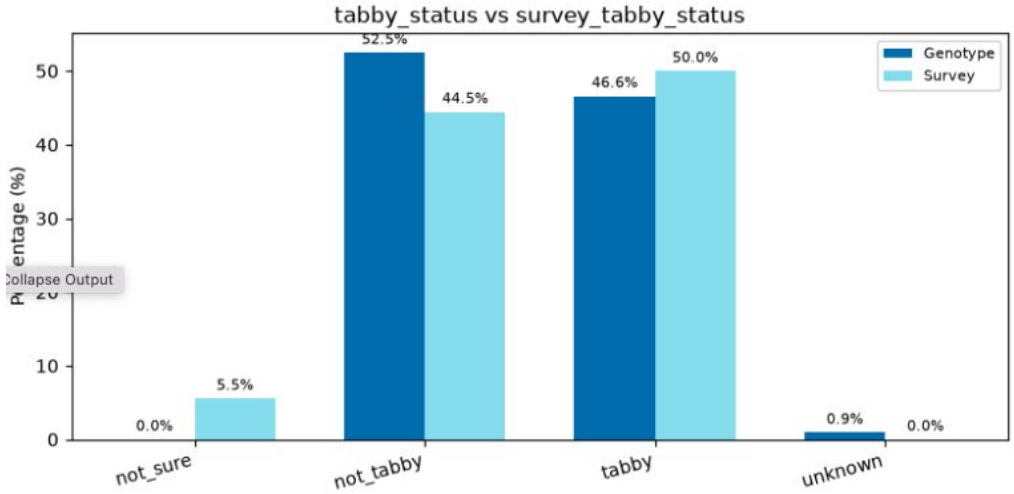
Tabbies have M-shaped marking on their forehead, stripes by their eyes and along their back, legs, and tail, and some have patterns on the body. Please just consider the non-white portions of CAT's coat.

- **Yes**
- **No**
- **Not sure**



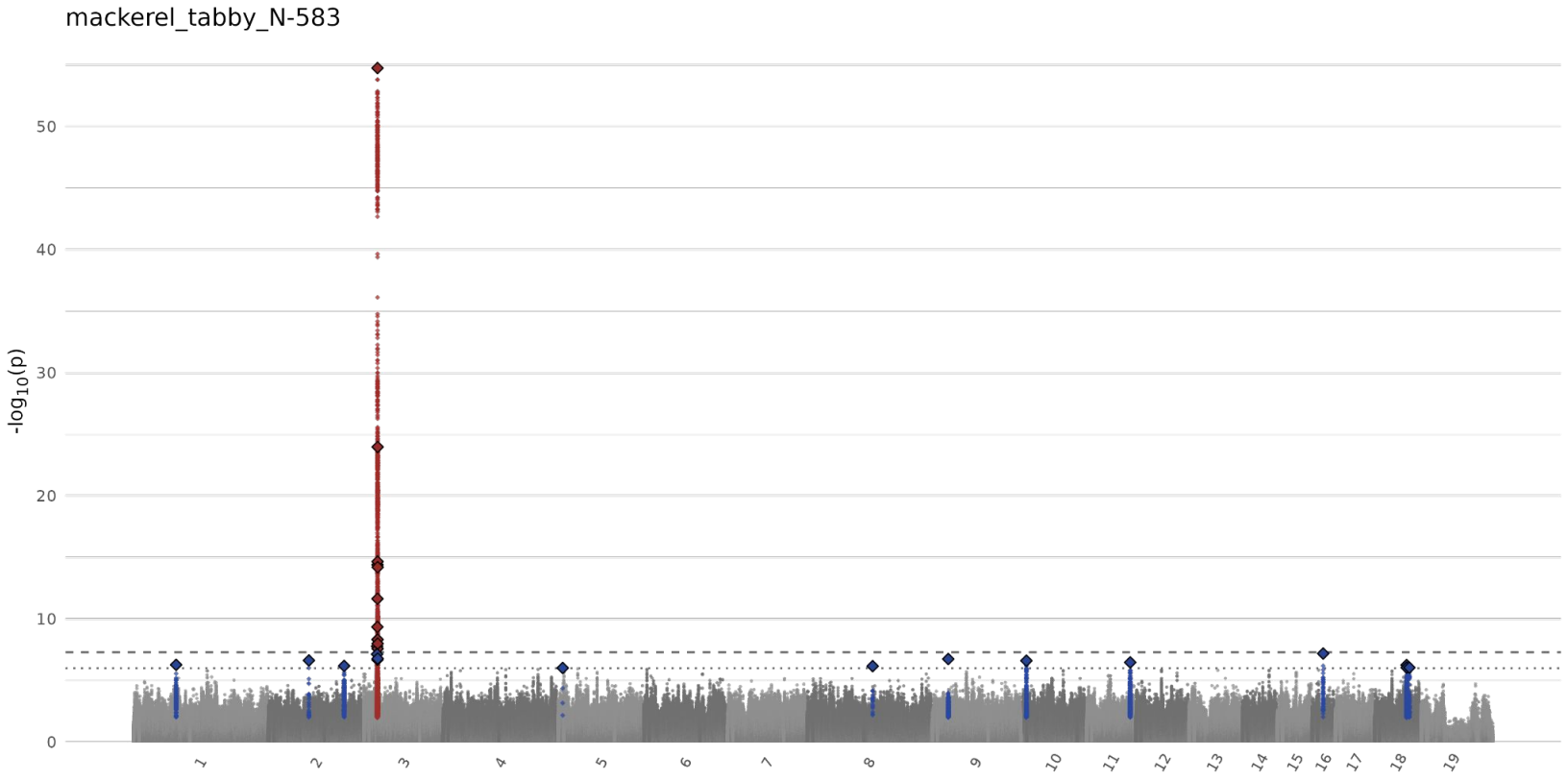
Distinctive M-shaped marking on forehead

Tabby status

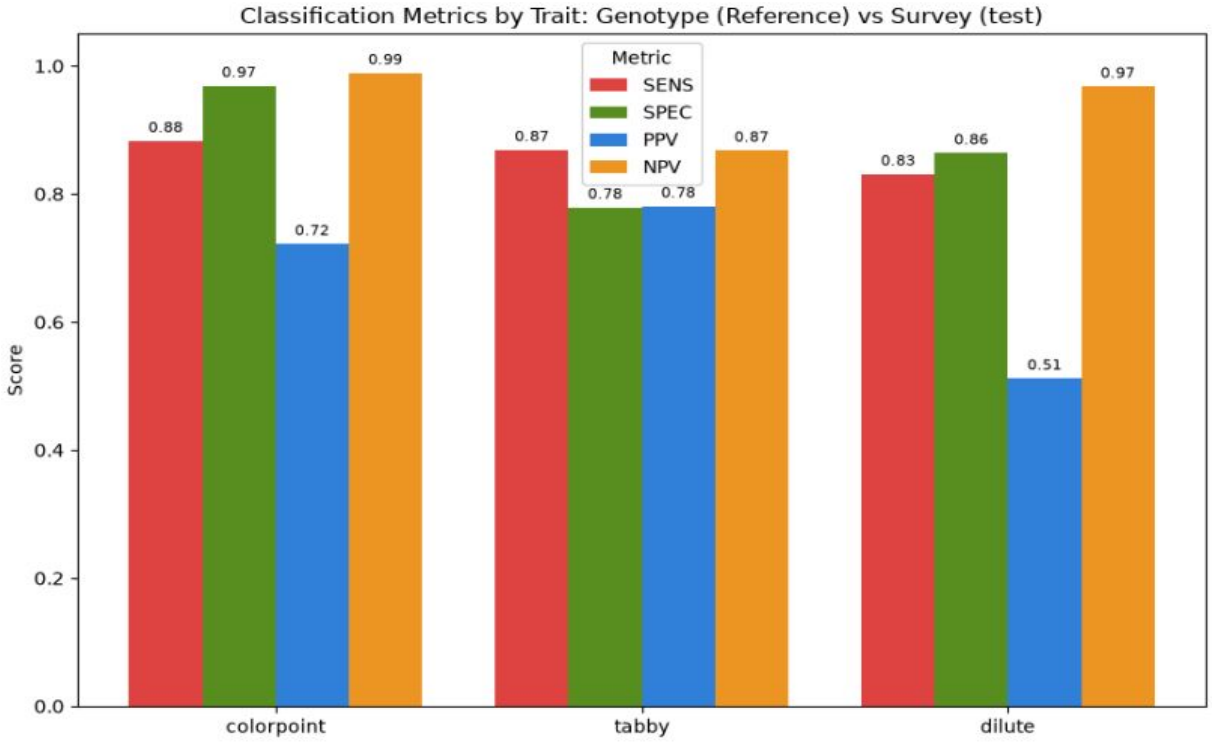


Overall discordance: 17.8%

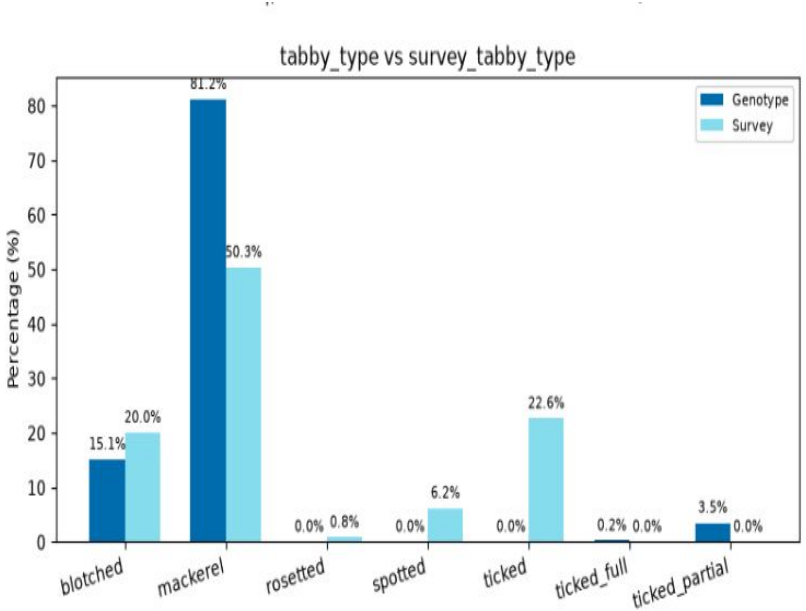
Darwin’s Ark tabby GWAS



Metrics for colorpoint, agouti, dilute

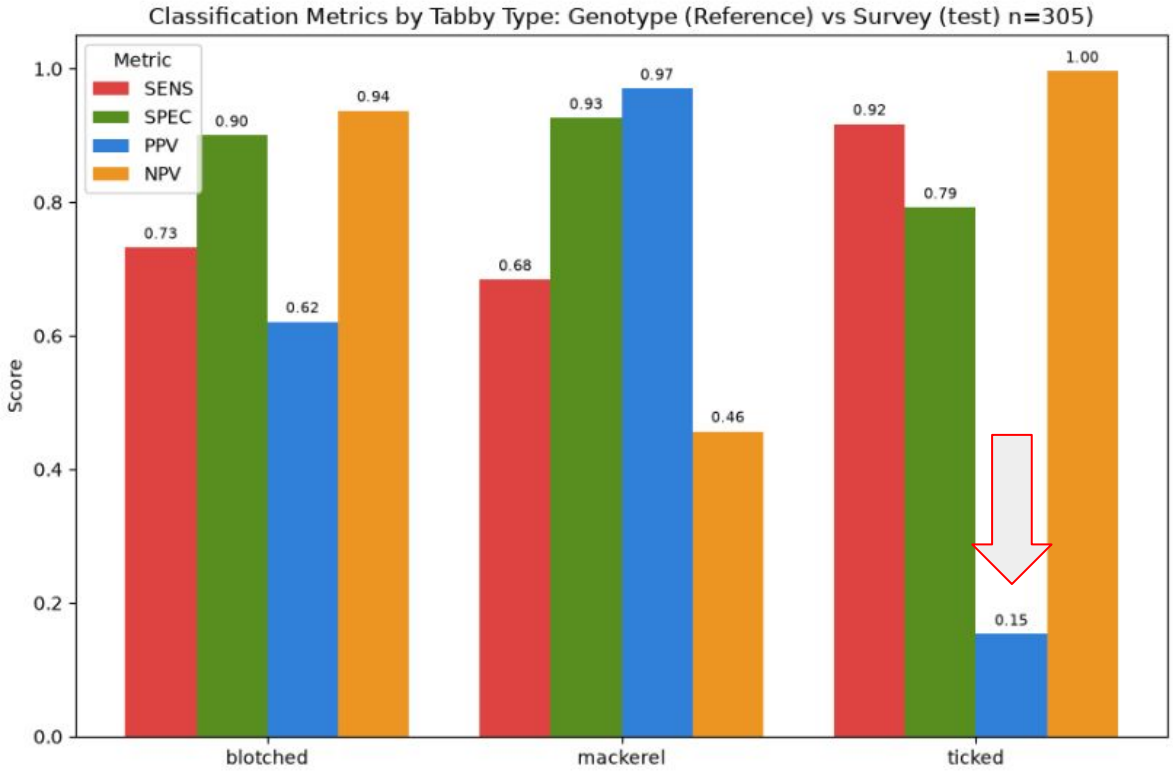


Tabby type discordance

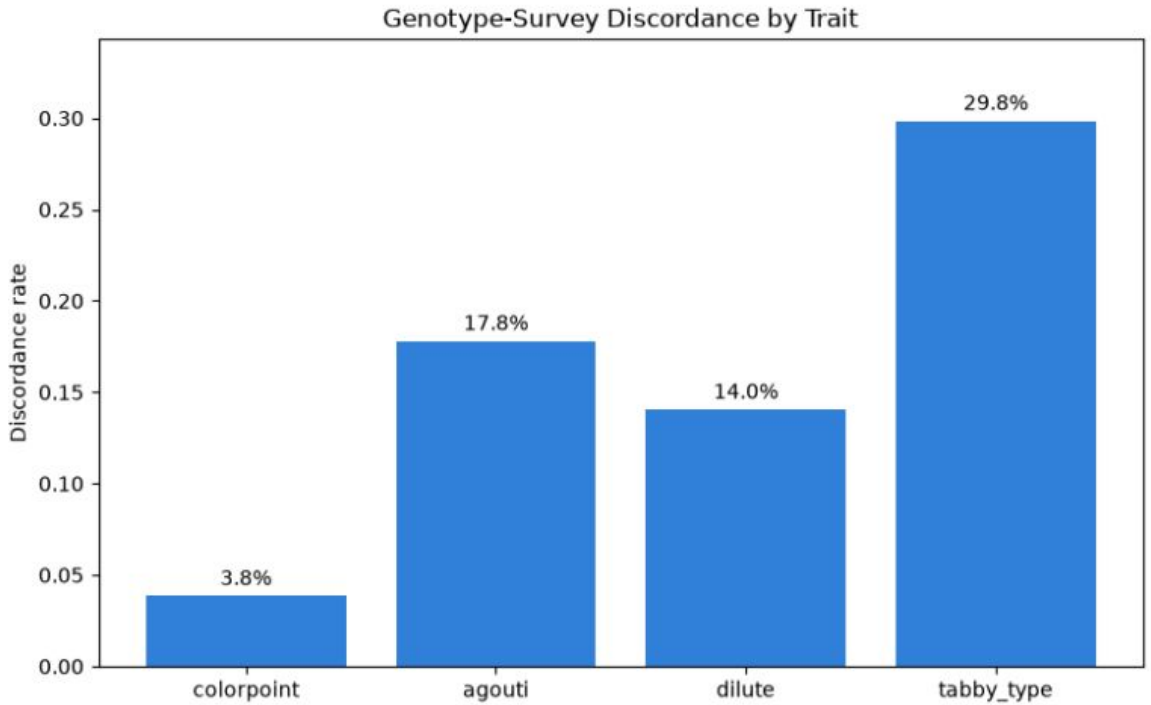


Overall discordance: 29.8%

Metrics for tabby type



Genotype-Survey Discordance

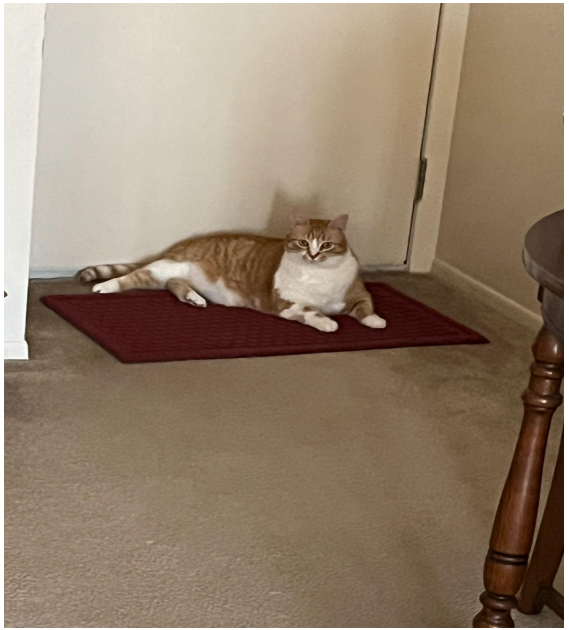


What does the Darwin’s GWAS have to say . . .?

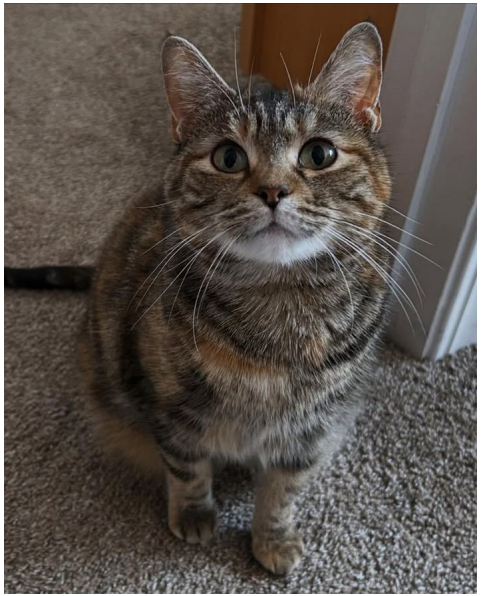
Coming soon

Diving in to the Ticked discrepancy

- 61 owners called their cat ‘Ticked’ when the genotype said ‘Mackerel’ or ‘Blotched’
- Owner misclassification ⇒



Mackerel



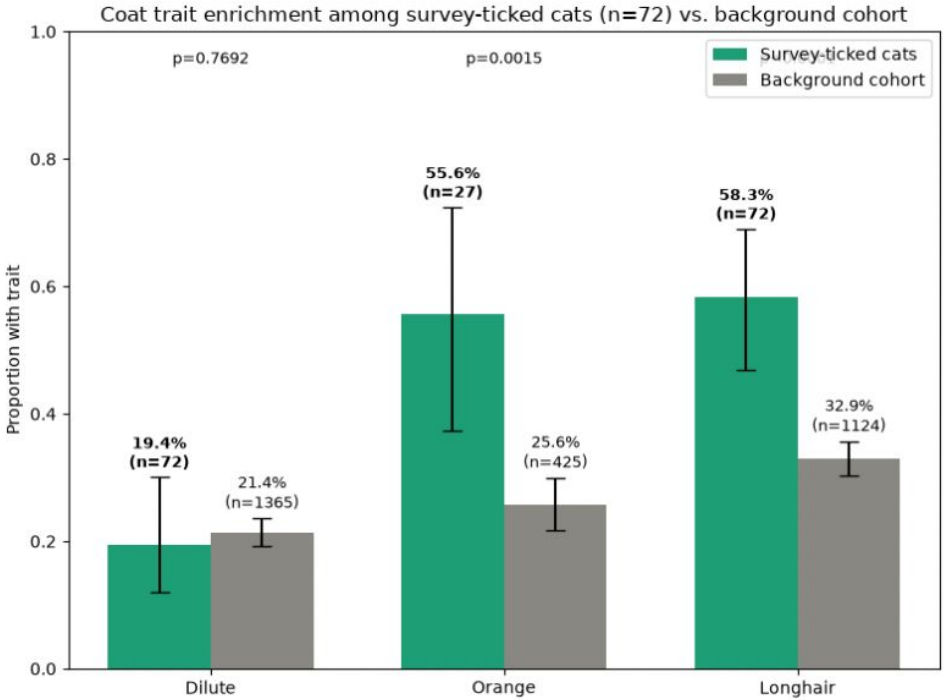
Classic (“Blotched”)

Diving in to the Ticked discrepancy

- Orange cats

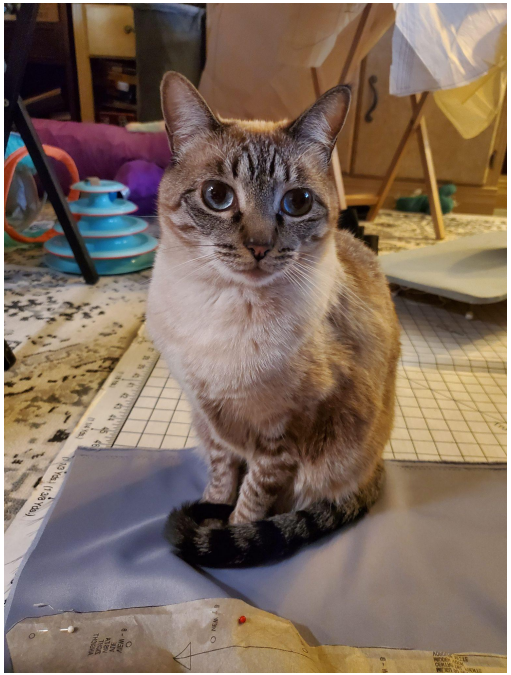


Ticked cats are enriched with risk factors



Diving into the Ticked discrepancy

- Some tougher calls



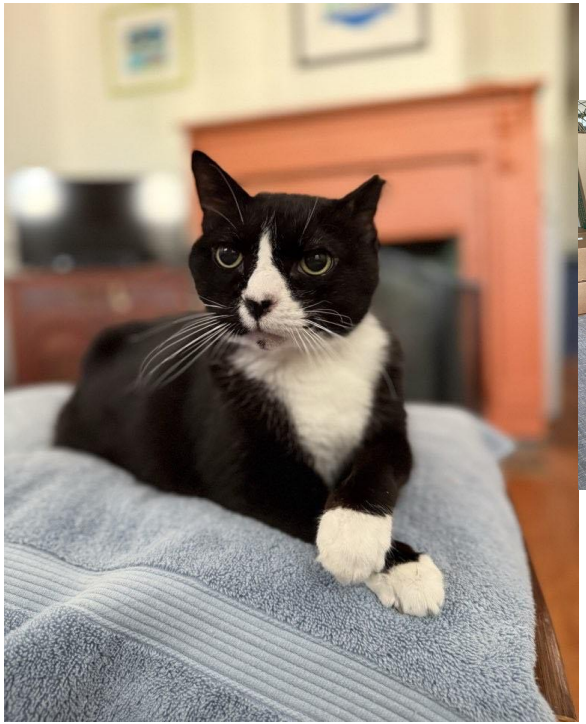
Colorpoint mackerel

Discordance in general

Another way to compare: Sets of traits from genotype and from survey

	geno_terms	survey_terms
0	((pattern_type, mackerel), (pattern_presence, tabby), (dilute, dilute), (colorpoint, not_colorpoint))	((pattern_presence, tabby), (colorpoint, not_colorpoint), (pattern_type, ticked))
1	((pattern_type, mackerel), (pattern_presence, tabby), (dilute, dilute), (colorpoint, not_colorpoint))	((pattern_type, mackerel), (pattern_presence, tabby), (dilute, dilute), (colorpoint, not_colorpoint))
2	((pattern_type, mackerel), (pattern_presence, tabby), (colorpoint, not_colorpoint), (dilute, not_dilute))	((pattern_type, spotted), (pattern_presence, tabby), (colorpoint, not_colorpoint), (dilute, not_dilute))

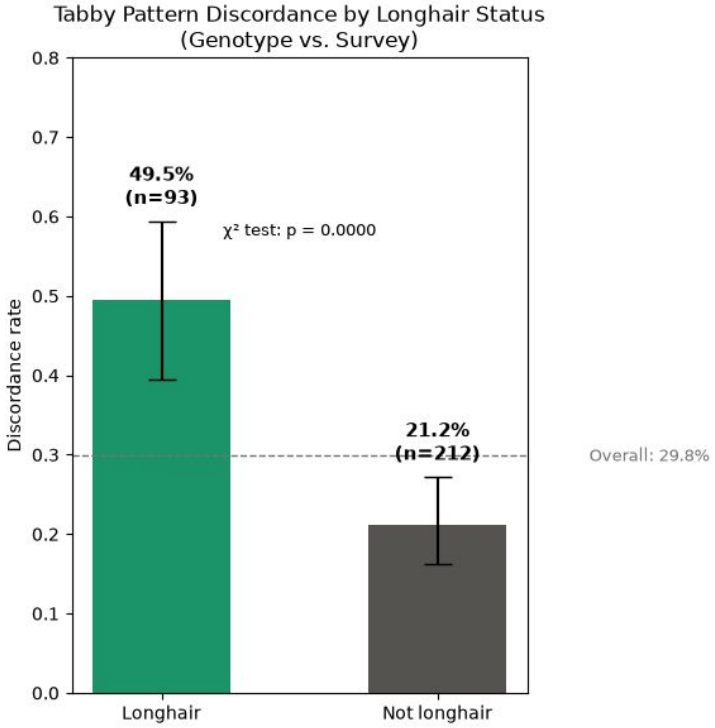
Total agreement (Easy cats)





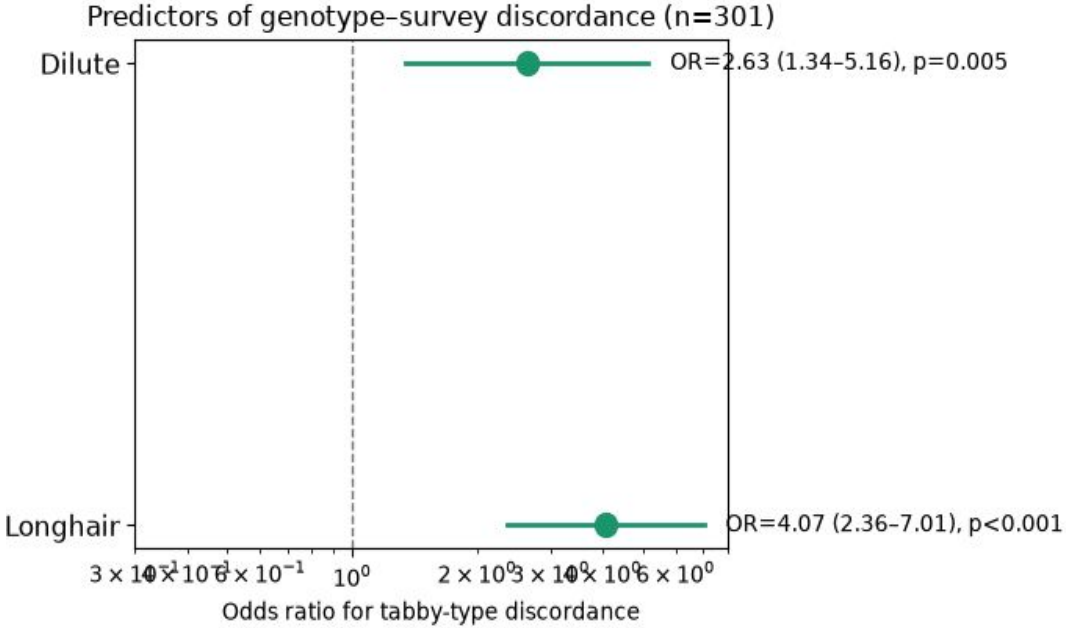


Longhair coat length predicts genotype-survey discordance



(in 305 cats with tabby type and survey tabby type not missing)

Longhair cats have 4.1× the odds of genotype-survey discordance



(in 305 cats with tabby type and survey tabby type not missing)

Risk factors: “Longhair”, “Dilute”, “Orange”

n_risk_factors	mean	count
0	0.146853	143
1	0.410072	139
2	0.545455	22
3	1.000000	1

Discordance increases with additional risk factors.

Two mismatches (Difficult cats)



GENOTYPE: Not dilute, not colorpoint, mackerel tabby

SURVEY: Dilute, not colorpoint, ticked tabby

Key Takeaways

- Survey takers may need more help answering some of the toughest questions on cat coat description (e.g., example photos)
- Darwin’s Ark has replicated and validated much of the literature on cat coat phenotypes
- Darwin’s Ark will be crucial to validate tabby patterns as well as to explore them further
- The Darwin’s Ark model works!